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Resetting autoimmune disease with CAR cell therapies

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Pathogenic B cell activation underlies many autoimmune diseases (AIDs), and their depletion is an attractive therapeutic approach. Chimeric antigen receptor (CAR)-expressing cells—initially developed and successfully used to treat certain cancers—are increasingly being developed to selectively deplete B cells and 'reset' the immune system in AIDs. In this Review, we survey this fast-developing field, providing insights on the current unmet needs in the treatment of AIDs and how CAR T cells could address these needs. In particular, we explore the concept of deep B cell depletion, discuss the currently available technologies and review the key targets (CD19 and B cell maturation antigen) relevant for the treatment of AIDs. We summarize current evidence on the efficacy, safety, risks and limitations of autologous and allogeneic CAR T cells in this setting. Finally, we discuss the future outlook—from a technological and clinical standpoint—for development of engineered CAR-expressing cell therapies for AIDs.

Pathological B cell activation has a central role in AIDs. This concept is supported by the observation that autoantibodies typically emerge years before the onset of clinical AID[1] and that B cell-depleting monoclonal antibodies can control AID activity[2]. B cells can trigger AIDs by three key mechanisms (Fig. 1): direct autoantibody-driven interference with physiological processes (such as the killing of thrombocytes by cell-specific autoantibodies in immune thrombocytopenia (ITP)); the formation of immune complexes that trigger inflammation (as happens in systemic lupus erythematosus (SLE)); and T cell activation, whereby B cells can act as antigen-presenting cells[3], leading to effector T cell activation and cell and tissue damage (as in type I diabetes mellitus). B cells have even been recognized to contribute to ulcerative colitis, traditionally considered to be predominantly T cell dependent[4,5]. Notably, B cells are also a rich source of cytokines, thereby contributing to induction and potentially also to resolution of inflammation[6]. The latter effect is associated with regulatory B cells that can suppress T cell activation and produce the immunoregulatory cytokine interleukin 10 (IL-10)[7]. Based on these multiple angles by which B cells orchestrate AIDs, their targeting is considered as an attractive therapeutic approach.

Treatment of AIDs has substantially improved owing to the introduction of small-molecule-based and protein-based targeted therapies[8]. In addition, monoclonal antibody-mediated B cell depletion has shown efficacy in the treatment of AIDs, further highlighting the role of B cells[2,9]. Nevertheless, current treatment approaches for AIDs are based on long-term immunosuppression to mitigate organ damage, not at eradicating the disease itself; cure is not achieved by such medications. While chronic administration of drugs interfering with immune function is feasible, it comes with substantial long-term drug and healthcare costs, and risks of enhanced infection rates and other drug-related toxicities. Hence, the key frontier in AID medicine today is to cure these diseases, ideally using a time-limited intervention that allows a 'reset' of the pathological immune system to a new state of immune tolerance. Autologous stem cell transplantation in patients with severe and treatment-refractory forms of AID has previously followed this concept with success[10]. However, this procedure can be complicated by substantial toxicities owing to the necessity of a harsh chemotherapy-based conditioning regimen that ensures full engraftment of the cells and the development of a 'new' immune system.

Modern technologies allow genetic engineering of therapeutic cells for selective deep B cell depletion, pushing the current limits of AID treatment. In this Review, we will discuss the unprecedented potential of CAR-expressing cells in the treatment of AIDs. These cells

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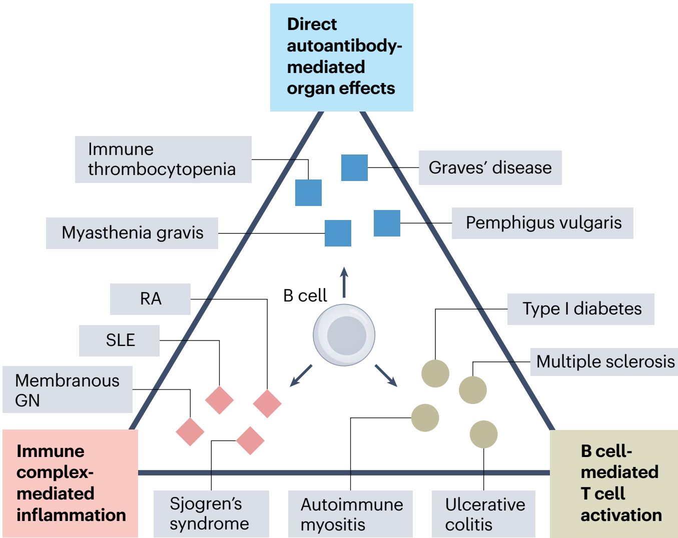

Fig. 1 | Contribution of B cells in AIDs. B cell-mediated pathology in AIDs can be driven by three key mechanisms, shown at each edge of the triangle. The top edge shows direct autoantibody-driven pathology such as lysis of thrombocytes by autoantibodies in immune thrombocytopenia, interruption of tight junctions in the skin and neuromuscular transmission by autoantibodies in pemphigus vulgaris and myasthenia gravis, respectively, or autoantibody-mediated thyroid gland stimulation in Graves' disease. Such processes are not typically associated with inflammation but rather constitute direct cell and organ damage triggered by autoantibodies. The left edge shows immune complex formation and inflammation. In this context, which is found in rheumatoid arthritis (RA), SLE, membranous glomerulonephritis (GN) and Sjogren's syndrome, immune complexes deposit in the affected tissues and trigger complement activation, which attracts myeloid-derived effector cells that trigger local inflammation and tissue destruction. In this cluster of AIDs, inflammation is more pronounced than in the first cluster. The right edge shows AIDs in which the effector function of autoantibodies is less established, but T cells seem to have a more central role as effectors. B cells are potent antigen-presenting cells, and thereby activate T cells, which then act as effectors leading to cell and tissue damage. Type I diabetes mellitus, autoimmune myositis, multiple sclerosis and ulcerative colitis are examples of such diseases.

can eliminate the entire B cell population and thereby allow for the establishment of a new state of immune homeostasis that is accompanied by sustained drug-free remission and potentially even cure.

Characteristics of CAR-expressing cells relevant to AID

CARs are engineered proteins with antigen binding and effector functions, which are introduced into T cells or natural killer (NK) cells where they are expressed on the cell surface (Fig. 2). Specific binding of the target antigen (CD19 on B cells or B cell maturation antigen (BCMA) on plasma cells) by the CAR is accomplished by an extracellular single variable immunoglobulin chain, while its effector function is mediated by an intracellular CD3-zeta and co-stimulation domain (4-1BB or CD28) that activates T cells and their killing function[11,12]. A strength of CD19-CAR T cell therapy is that a single infusion of engineered cells may completely deplete B cells from the body. Given that only a limited number (usually less than 100 million cells) of autologous CAR T cells are infused, this Herculean mission of total B cell depletion is only possible because CAR T cells proliferate upon antigen recognition in vivo (homeostatic proliferation) and thereby multiply their force—a phenomenon known as the 'living drug effect'. Furthermore, each CAR T cell eradicates multiple target cells[13]. Finally, CAR T cells have access to all organs of the human body and, therefore, efficiently deplete B cells across the many target tissues and secondary lymphatic organs impacted by AIDs, including the central nervous system and brain.

Two key targets are currently being investigated with CAR T cell therapy in AIDs: CD19 and BCMA. CD19-CAR T cells target a large part of the B cell lineage comprising pro-B cells, pre-B cells and intermediate B cells in the bone marrow as well as naive B cells, memory B cells and plasmablasts in the circulation, secondary lymphatic and respective target organs. CD19-CAR T cells lead to a complete extinction of the B cell lineage with the only exception being CD19-negative plasma cells[14]. CD19 constitutes an excellent target with virtually exclusive expression on B cells. Biopsy studies of the lymph nodes and target organs in patients with AID, having received CD19-CAR T cells, have shown complete depletion of B cells and plasmablasts in all tissues investigated. The other key target antigen is BCMA, which was originally thought to be only expressed on plasmablasts and plasma cells but seems to also be expressed on B cells at lower levels. BCMA is used as a target of CAR T cells targeting multiple myeloma, a malignant plasma cell disorder, and can therefore also be used to deplete autoimmune plasma cell clones[15]. It is important to note that targeting CD19 and targeting BCMA are biologically distinct therapeutic approaches: the first depletes naive and memory B cells and plasmablasts, leaving long-lived plasma cells untouched, while the latter also eliminates long-lived plasma cells.

Feasibility of generating functional CAR T cells from individuals with AID

Manufacturing sufficient numbers of functional CAR T cells is essential for successful cell therapy. In individuals with AID, disease-associated lymphopenia (that is, in SLE), impairment of T cell function due to immunosuppressive drug therapies and enhanced T cell exhaustion due to chronic T cell stimulation have caused substantial concern with regard to manufacturing sufficient numbers of therapeutic cells. So far, however, none of these potential concerns has been substantiated. Sufficient numbers of autologous T cells can be retrieved ($>1\times10^8$), transduction efficacy is good (>30%) and ex vivo expansion excellent (>50-fold), allowing for production of a high surplus of cells and enabling cryopreservation of additional bags for potential later infusions[16,17]. Usually, among T cells expressing the CAR, $CD4^+$ cells prevail over $CD8^+$ cells and strong enrichment of $CD27^-CD45RA^-$ effector memory T cells is observed, while exhaustion markers such as CD57 and PD-1 are decreased[16,17]. Overall, the production of CAR T cells from individuals with AID is consistently robust despite the variability in diseases and pretreatments. Production failures are generally rare.

Furthermore, while cessation of immunosuppressive treatment regimens before leukapheresis and T cell harvesting has initially been recommended to ensure good viability and expansion of the cells, the robustness of manufacturing suggests that pausing immunosuppression for leukapheresis may not be needed, and a shorter pause before CAR T cell infusion may be feasible (and even beneficial, as it avoids disease flares). As the different immunosuppressants and immunomodulators vary substantially in their mode of action (for example, T cell targeting versus non-T cell targeting), pharmacological structure (for example, small molecule versus monoclonal antibody) and half-life, the protocols for pausing immunosuppressants should be tailored to the specific compound. In addition, treatment with low to moderate doses of glucocorticoids has not shown any effect on CAR T cell production.

Notably, various production processes are used that differ in the duration of ex vivo T cell expansion and in whether point-of-care decentralized manufacturing or central manufacturing is performed. Newer methods that shorten the ex vivo expansion time of CAR T cells before product administration aim to reduce the turnaround time (from cell retrieval until administration) of centrally manufactured CAR T cell products. In addition, point-of-care manufacturing is another emerging method, where the cells are produced in local good manufacturing practice-certified facilities and can be administered with no need to freeze or transport the product, allowing a fresh-to-fresh product circle with short turnaround time.

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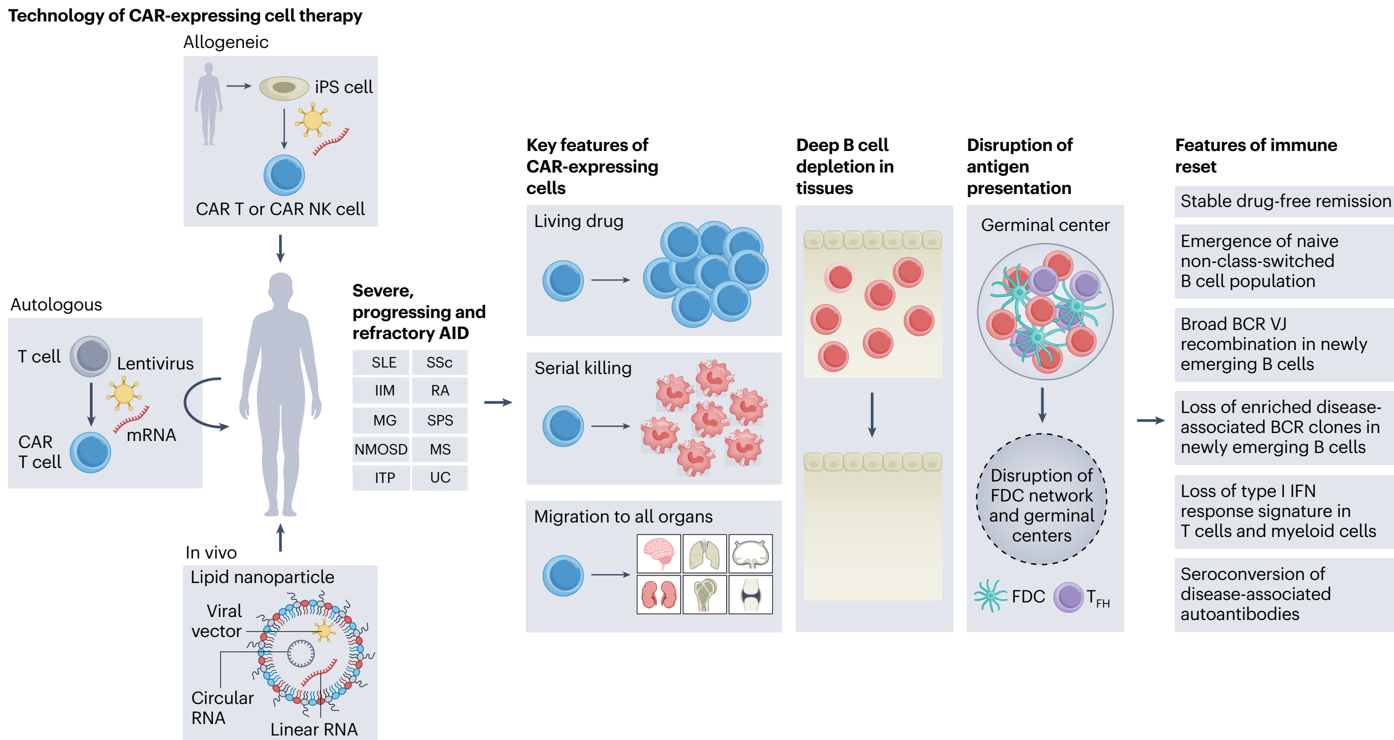

Fig. 2 | CAR-expressing cell therapy in AIDs. Left, treating patients who have AID with CAR-expressing cells utilizes three key technologies and approaches: infusion of autologous CAR-expressing cells; allogeneic CAR-expressing cells; or lipid nanoparticles containing RNA or viral vector encoding the CAR (in vivo CAR T cell therapy). The box beside the patient shows key indications in which these CAR-expressing cells were tested. Center/right, CAR-expressing cells are 'living drugs' with expansion potential, which mediate deep B cell depletion and disruption of germinal centers in lymph nodes, triggering immune reset characterized by various molecular and clinical features. FDC, follicular dendritic cell; IFN, interferon; IIM, idiopathic inflammatory myopathy; MG, myasthenia gravis; MOA, mode of action; MS, multiple sclerosis; T_{FH}, follicular helper T cell; UC, ulcerative colitis.

Efficacy of autologous CAR T cell therapy in AID

As CAR T cell therapy is a complex procedure, careful selection of patients with AID is required. Independent of disease indication, patients with severe and treatment-resistant AID are suited to CAR T cell therapy, in line with the concept of CAR T cell therapy in oncology to treat resistant patients. Notably, objectives of CAR T cell treatment of severe and treatment-resistant AID can be the prevention of death (for example, in case of rapid-progressive interstitial lung disease or cardiac involvement in systemic sclerosis (SSc) and myositis), organ failure (for example, end-stage kidney disease in SLE, loss of insulin-production beta cells in type 1 diabetes mellitus) and permanent disability (for example, paralysis in multiple sclerosis, loss of joint and muscle function in rheumatoid arthritis and IIM). Rapidly progressing disease may require fast decisions to use CAR T cell therapy within a window of opportunity that allows prevention of organ failure. Specific risk constellations that require rapid action include MDA5-autoantibody positivity in patients with IIM, rapid-progressive glomerulonephritis in SLE, early diffuse skin pattern and signs of inflammation in SSc, myasthenic crises in myasthenia gravis and progressive disease course in multiple sclerosis, to name a few.

Early preclinical data in lupus-prone mice suggested efficacy of B cell-directed CAR T cells in SLE[18], prompting the first clinical use of CAR T cell therapy in a patient with AID. This first patient was a 20-year-old woman with severe, treatment-refractory SLE involving the kidneys, the heart, lungs, skin and joints. The treatment, carried out in March 2021 (ref. 19), was composed of autologous CAR T cells that were manufactured by transduction of the cells with a second-generation lentiviral vector encoding a CD19-CAR (zorpocabtagene autoleucel). The target binding moiety of this CAR is a single chain variable fragment of immunoglobulin (scFv) derived from a murine anti-human CD19 hybridoma clone FMC63 (ref. 20) with a 4-1BB co-stimulation domain. This first case uncovered several key aspects of autologous CAR T cell therapy that were consistently reproduced in further studies. First, the production of CAR T cells from patients with AID is highly stable. Second, AID rapidly improves following CAR T cell treatment, with major clinical responses seen across different diseases. Third, the treatment procedure is relatively safe, with lower incidence of cytokine release syndrome (CRS), immune effector cell-associated neurotoxicity syndrome (ICANS) and immune effector cell-associated hematotoxicity (ICAHT) than observed in hemato-oncology. Finally, immunosuppressive treatment regimens can be successfully stopped following CAR T cell therapy in most patients.

Since this first case, preliminary efficacy data on CAR T cell treatment have been retrieved in a large variety of diseases including SLE[16,17,21–25], SSc[17,26–28], IIM[17,29], rheumatoid arthritis[30,31], granulomatosis with polyangiitis (GPA)[32], multiple sclerosis[33,34], myasthenia gravis[35–39], neuromyelitis spectrum disorder (NMOSD)[40], stiff person syndrome (SPS)[41], autoimmune encephalitis[42], ulcerative colitis[43] and ITP[44,45] (Fig. 2). Notably, pediatric patients with SLE, IIM and SSc have also been successfully treated with CAR T cells[46–50]. These data are based on individual case reports, case series and early clinical studies and representing only a fraction of patients with AID currently treated with CAR T cells—as a plethora of late clinical studies are currently ongoing.

The first phase 1/2 study of CD19-CAR T cell therapy in patients with AID showed safety and efficacy in patients with SLE, IIM and SSc[51]. Most patients with AID who have been treated with CAR T cell therapy received cells targeting CD19 and most clinical studies (including pivotal studies) are based on CD19-CAR T cell therapy. Among CD19-CAR T cell products, rapcaptagene autoleucel and zolacabtagene autoleucel (both both 4-1BB co-stimulation-based with fast production method), resecabtagene autoleucel, obecaptagene autoleucel, relmacabtagene autoleucel, zorpocabtagene autoleucel (all

4-1BB co-stimulation-based) and mivocabtagene autoleucel (CD28 co-stimulation-based) are currently entering or have entered phase 2 studies in AIDs including SLE, IIM, SSc, rheumatoid arthritis, GPA, multiple sclerosis, myasthenia gravis and SPS. Also, CAR T cell therapies targeting BCMA[33,38,40], CD19 and BCMA[21,23,36,37], and CD19 and CD22 (ref. 52) have been tested in patients with AID, showing good safety and efficacy. These approaches are intended to achieve an even more comprehensive B cell depletion (including plasma cells, which express BCMA) than CD19-directed CAR T cells.

Further research is needed to determine whether BCMA targeting provides efficacy or safety advantages over CD19-targeting in specific groups of patients with AID. It can be assumed that in some patients, particularly in those with high autoantibody titers despite previous B cell depletion, autoimmunity may be anchored more so in the plasma cell compartment and BCMA targeting may be needed to eradicate this form of autoimmunity. On the other hand, BCMA targeting inevitably leads to hypogammaglobulinemia, which requires substitution with IgG and may induce long-term humoral immune impairment. Several BCMA-directed CAR T cell products, such as equecaptagene autoleucel and Descartes-08 (mRNA-based transfection) are currently being tested in pivotal phase 2 studies in NMOSD, myasthenia gravis and SLE.

Efficacy of CAR T cell therapy in AID can be monitored by clinical, cellular and biochemical readouts[53,54]. Clinical readouts are disease dependent and usually comprise composite scores that measure clinical remission (for example, in SLE and rheumatoid arthritis), relevant clinical responses (for example, in IIM and SSc) or lack of progression (for example, in multiple sclerosis). Important cellular readouts for efficacy include the dynamics of CAR T cell expansion and decline in vivo and the completeness and duration of B cell depletion. Biochemical endpoints relevant to efficacy monitoring of CAR T cell treatment are the regression of acute-phase responses, the normalization of organ-specific markers (for example, proteinuria in kidney disease; troponin I level in heart disease; creatinine kinase in muscle disease) and the decline or seroconversion of disease-associated autoantibodies. Given the complexity of the treatment, high-level and sustained responses are preferred (see Table 1 for a list of key disease-related outcomes).

The central aim of CAR T cell treatment in AID is to reset the immune system and induce a state of sustained drug-free remission. Hence, immunosuppressive drug therapy is usually stopped before the CAR T cell infusion. Relapses can occur, although they seem to affect only a minority of patients. Nonetheless, they constitute a central unresolved issue as no predictors of relapse after CAR T cell treatment have yet been identified. Potential mechanisms of relapse include incomplete tissue depletion of B cells, persistence of CD19-negative plasma cells, survival of clones in protected niches (such as lymph nodes), antigen-driven reseeding, waning CAR persistence and anti-CAR immune responses. Relapses usually occur 9–24 months after CAR T cell treatment and manifest with the onset and continuous worsening of disease-specific symptoms. Repeated CAR T cell treatment has been described, whereby the switch to another CAR T cell product was necessary owing to a T cell reaction against the original CAR, thereby neutralizing its efficacy and preventing CAR T cell expansion[55]. In this context, rescue therapy of relapsing AID by BCMA-CAR T cells has been described[55].

Table 1 | Key disease-related outcomes in patients with AID treated with CAR-expressing cells

Indication	Overarching outcomes	Disease-specific outcomes
Systemic lupus erythematosus	Reversible complete peripheral and tissue B cell depletion AND drug-free state >6 months without signs of disease activity	DORIS remission fulfilled; LLDAS (lupus low disease activity state) fulfilled
Systemic sclerosis		Revised ACR-CRISS (ACR Composite Response Index in Systemic Sclerosis) fulfilled; EUSTAR-AI (European Scleroderma Trials and Research Group activity index) < 2.5
Idiopathic inflammatory myopathy		ACR moderate/major response[5] fulfilled
Rheumatoid arthritis		ACR/EULAR remission fulfilled; disease activity score (DAS) 28 < 2.6
Myasthenia gravis		MG-ADL (Myasthenia Gravis Activities of Daily Living) scale ≤ 1; quantitative myasthenia gravis (QMG) score ≥ 3-point reduction
Neuromyelitis optica spectrum disorder		Prevention of relapse
Multiple sclerosis		No progression and improvement of EDSS (Expanded Disability Status Scale)
Stiff person syndrome		Improvement of timed 25-foot walk test
Granuloamatosis with polyangiitis		Birmingham vasculitis activity score = 0
Ulcerative colitis		SCCAI (Simple Clinical Colitis Activity Index) < 2.5
Immune thrombocytopenia		Normal platelets counts (>150 g l^{-1})

Data on allogenic CAR-expressing cell therapy in AID

Allogeneic CAR T cells offer an attractive alternative to autologous CAR T cell products owing to their immediate availability, manufacturing scalability and reliable quality. However, they must overcome the challenges of graft-versus-host disease (GvHD) and host-versus-graft rejection. Very recently, a burgeoning trend has emerged in exploring allogeneic B cell-targeting cellular therapies for autoimmune disorders. These therapies have achieved promising clinical remission in patients with severe, refractory diseases, including SSc[56,57], SLE[56–58] and IIM[56]. A primary strategy to mitigate GvHD toxicities mediated by αβ T cell antigen receptors (TCRs) recognizing mismatched human leukocyte antigen (HLA) involves disrupting αβTCR expression, such as by knocking out the *TRAC* gene using CRISPR–Cas9 (refs. 56,58,59). An alternative approach is to use effector cells naturally lacking αβTCR, like donor[60] or induced pluripotent stem (iPS) cell-derived NK cells[57]. Both strategies have proven effective, with minimal GvHD events reported[56–58].

A key challenge for the in vivo persistence of allogenic cells is host T cell-mediated rejection via incompatible HLAs. Conversely, the complete absence of HLAs can trigger clearance by host NK cells. To create hypoimmunogenic cell therapy products, multiple strategies have been adopted. One method involves knocking out major HLAs responsible for rejections (HLA-A/B and all HLA-II) while preserving minor ones to avoid activating host NK cells[56,58,59]. Adopting this strategy, allogeneic anti-CD19-CAR T cells from a HLA-mismatched background achieved variable in vivo expansion, peaking at 1–3 weeks after infusion. Despite a more than 10-fold variance in peak expansion levels, all treated patients—six receiving the product TyU19 (refs. 56,59) and five receiving YTS109 (ref. 58)—achieved deep clearance of circulating B cells, which persisted for 1–2 months before B cells recovered.

Most transferred allogeneic CAR T cells fall below the detection range by flow cytometry within the first month[56,58]. Notably, existing data suggest that prolonged CAR T cell presence and higher expansion do not correlate with prolonged B cell depletion. $CD19^+$ B cells were

Table 2 | Toxicities related to treatment with CAR-expressing cells in patients with AID

Toxicity	Onset	Grading	Prevalence	Treatment
CRS	Acute	I–IV[a]	Frequent	Tocilizumab
ICANS	Subacute	I–IV[a]	Rare	Glucocorticoids
ICAHT	Subacute	I–IV[b]	Rare	G-CSF
Late neutropenia	Delayed	I–IV[b]	Rare	G-CSF
LICATS	Subacute	I–IV[c]	Frequent	Glucocorticoids
Immune effector cell-associated hemophagocytic lymphohistiocytosis-like syndrome (IEC-HS)	Subacute	-	Rare	Various regimens[e]
Hypogammaglobulinemia	Delayed	-	(In)frequent[d]	IVIG
Infections	Subacute	-	Infrequent	Acylovir, cotrimoxazol[f]

[a]ACTST grading. [b]EHA/EBMT grading[62]. [c]Grading according to ref. 65. [d]Dependent on CAR target: obligatory with BCMA-CAR, infrequent with CD19-CAR. [e]Glucocorticoids, IL-1 receptor antagonist (anakinra), etoposide, JAK inhibitors (ruxolitinib); anti-IFNγ monoclonal antibody (emapalumab). [f]Prophylactic treatments.

shown to be restored at around 1–2 months for all patients with AID, who received allogeneic CAR T cells. Clinical remission was achieved for nearly all patients within the first 3 months, which was largely maintained without additional immunosuppressants throughout the 6-month monitoring period[56,58,59], except for one patient with minor fluctuation[59]. Furthermore, clearance of tissue-resident B cells and reversal of challenging pathologies (for example, fibrosis) in multiple organs and of proteinuria were observed in multiple patients receiving allogeneic CAR T cell treatments[56,58,59].

Another strategy to reduce allogenic rejection involves disrupting all HLA expression and then circumventing host NK cell activation by transducing HLAs with minor allogeneic incompatibility, as has been adopted for iPS cell-derived CAR NK cells[57]. However, owing to a lack of in vivo expansion, both hypoimmunogenic and unmodified donor-derived CAR NK cells persisted poorly and required repetitive dosing to compensate. For instance, Wang et al.[57] administered a course of four doses of 600 million CAR NK cells within 10 days, whereas Gao et al.[60] adopted up to three doses of 1.5 billion cells per dose within 9 days. Both approaches achieved depletion of circulating B cells for approximately 3–4 weeks, despite the brief persistence of infused NK cells. Patients showed continued improvement to varying degrees, which was largely maintained throughout the monitoring period with a minimal to moderate steroid use.

All treatments reported a desirable safety profile with manageable toxicities[56,58–60]. Although therapeutic safety and efficacy must be validated in larger pivotal trials, hypoimmune allogeneic CAR T cells show the potential for efficacy comparable to autologous CAR T cells. For CAR NK cells, short persistence and a lack of expansion can be compensated by repeated dosing, although the optimal dosing regimen to achieve clinical efficacy comparable to CAR T cells remains to be explored. Overall, scalable, ready-to-use and homogeneous allogeneic B cell-targeting therapeutic cell products represent a compelling alternative to autologous CAR T cell treatments.

Safety of CAR-expressing cell therapy in AID

Safety concerns relating to CAR T cell (and CAR NK cell) therapy include the risk of CRS and ICANS that, in the treatment of B cell malignancies, can be severe[61,62]. CRS results from T cell activation upon target cell engagement and cytokine release by myeloid cells. Excessive production of IL-6 and IL-1β can lead to a cytokine 'storm' that can be life-threatening. ICANS, which appears to be based on cytokine leakage into the brain, can also be severe and requires timely and adequate therapeutic measures. Based on current knowledge of CRS and ICANS, monitoring and treatment strategies—such as IL-6R inhibition in CRS and glucocorticoid treatment in ICANS—have been established to control such events. There is also a risk of ICAHT linked to the lymphodepletion regimen administered before CAR T cell infusion[63]. Prolonged leukopenia and neutropenia increase the infection risk and must be avoided, especially because infection risk is already high in patients with AID. On the other hand, short leukopenia lasting only for a few days after lymphodepletion can be managed easily without substantial infection risk; it also fosters homeostatic proliferation of CAR T cells, which leads to higher therapeutic cell drug exposure and deeper B cell depletion.

Key toxicities of CAR T cell therapy in AID and their management are listed in Table 2. Remarkably, overall incidence of CRS is lower in AID than in oncology patients and severity of CRS is usually confined to mild events (CRS grade 1), manifesting with fever in most cases[64]. This low risk for higher-grade CRS in patients with AID is likely based on the overall lower B cell burden in nonmalignant compared to malignant diseases—and as a result, a less intense activation of the T cell through the CAR. Other factors that could in principle influence the risk of CRS in patients with AID are the number of CAR T cells administered, the production process used (for example, shorter ex vivo expansion, more robust in vivo expansion) and the co-stimulation domain of the CAR vector (for example, 4-1BB or CD28) that has been selected. CRS occurs within days after infusion of CAR T cells and is managed by antipyretic drugs (for example, acetaminophen) and the anti-IL-6 receptor antibody tocilizumab. ICANS only rarely occurs in patients with AID and is treated with glucocorticoids and, more rarely, with IL-1 inhibitor anakinra. Owing to the overall paucity of ICANS cases in patients with AID, no specific risk factors have yet been identified, although previous CRS or concomitant infection may be associated with the development of ICANS.

Hematotoxicity (ICAHT) is also rare in patients with AID treated with CAR T cells, with most of the patients showing no or only short (up to 2 weeks) leukopenia and virtually no thrombocytopenia, indicating an overall good bone marrow reserve in these patients[16,17]. Some cases of late neutropenia (>28 days after CAR T cell infusion) have been observed, with most cases being reversible upon administration of granulocyte colony-stimulating factor (G-CSF). The pathogenesis of delayed neutropenia is not fully clear, but it may represent a bone marrow manifestation of local immune effector cell-associated toxicity syndrome (LICATS; see below) or a transient alteration of the stem cell niche in the bone marrow. Levels of the stem cell factor SDF-1 have been shown to be related to the presence of late neutropenia[63]. SDF-1 regulates the development of hematopoietic stem cells, neutrophils and very early B cells in the bone marrow. The initial recovery of early B cells in the bone marrow may compete for SDF-1 and, therefore, impair neutrophil development and migration.

LICATS, which manifests as transient worsening of organ function—usually around the time of maximum expansion of CAR T cells or after—has also been described in patients with AID treated with

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CAR T cells[65]. It usually involves the organs previously affected by the respective AID, for example, worsening of rash or kidney function in SLE, increase of serum creatinine kinase levels in IIM or enhanced skin involvement in SSc. LICATS, in contrast to CRS, is not a systemic reaction but a local process, likely based on a local inflammatory response when CAR T cells are cleansing tissue-resident B cell infiltrates. As the bone marrow is rich in B cells and neutrophils, LICATS in the context of killing of bone marrow B cells could also contribute to neutropenia. Notably, LICATS should not be confused with disease relapse that, if it occurs, usually happens much later (>6 months after CAR T cell infusion), when CAR T cells have disappeared and B cells are reconstituted. A wait-and-watch strategy is preferred if LICATS is mild, but glucocorticoids can be used for more severe or prolonged cases.

Long-term safety considerations span infections, reproductive health, cardiovascular disease and malignancies. Regulators currently ask for a 15-year documentation of long-term toxicities of CAR T cells in AID. Long-term toxicity data in AID are scarce owing to the relative novelty of the treatment. As most patients experience rapid B cell reconstitution and stop immunosuppression and glucocorticoid treatments (which add to infection risk), it can be speculated that long-term infection risk after CAR T cell infusion may be lower than before the treatment. The first long-term data on CD19-CAR T cell treatment in AID with up to 3 years' follow-up of 15 patients show no elevated infection risk[17]. Whether this observation also pertains to BCMA-CAR T cell treatment remains to be shown. Specific attention should also be paid to late infection risk due to hypogammaglobulinemia and impaired vaccination responses.

Cardiovascular risk is generally increased in AID. In individuals with SLE, for instance, cardiovascular events are the major cause of death[66]. Systemic autoimmunity, being characterized by immune complex formation, complement activation and enhanced proinflammatory cytokine production, is associated with vascular inflammation. Therefore, it might be speculated that cardiovascular disease may be better controlled when achieving an immune reset and sustained drug-free remission—although this remains to be shown.

Reproductive health is crucial in individuals with AID as most patients are women of reproductive age. AID can complicate pregnancies and some immunosuppressive treatments are incompatible with pregnancy, making sustained drug-free remission desirable. Although more data are needed, several factors suggest that CAR T cell therapy may not be harmful for reproductive health of patients with AID. First, CAR T cells are only transiently present in patients with AID and are not expected to cross the placental barrier; second, the current type and dose of the conditioning regimen with cyclophosphamide and fludarabine used for CAR T cell therapies is not considered to impair female fertility[67]; and third, ovarian protection with gonadotropin-releasing hormone analogs can be used in case of decreased ovarian reserve. Meticulous documentation of pregnancies and the immune system of the offspring of patients with AID receiving CAR T cells will be needed to shed light on this area.

Finally, long-term risk for malignancies will need to be documented. T cell lymphoma has been reported in patients who received CAR T cells to treat B cell lymphoma[68]. While concerns have been raised about potentially oncogenic random integration of lentivirus into patients' DNA, detailed workups of some cases in oncology patients have shown that T cell clones with mutations in tumor-associated genes such as *DNMT3A* or *TET2* were already present before the treatment, which, upon insertion of the CAR, can expand[69]. Whether such risks also pertain to AID is currently unknown, but the fact that T cell lymphoma risk is intrinsically increased (100-fold) in patients with B cell lymphoma compared to the general population, and that the presence of CAR T cells is not stable in patients with AID, together makes it unlikely that patients with AID receiving CAR T cells have an increased risk for T cell lymphomagenesis[70].

Concept of immune reset, sustained drug-free remission and cure

The observation of a sustained and stable drug-free state in patients with AID receiving CAR T cell treatment led to the term 'immune reset' to describe a state of newly established immune tolerance that had been lost before the treatment, and is difficult to regain by conventional immunosuppressive treatment (Fig. 3). Immune reset refers to a new homeostatic immune system that can stably manage challenges like infections, vaccinations, tissue stress and damage, without triggering a relapse of AID. While the term 'immune reset' was prompted by clinical observations, more work is needed to characterize it in detail. Clinically, it involves a lack of disease activity despite absence of treatment. Absence of drugs over a period of more than 6 months after CAR T cell infusion, that is, in the case of B cell reconstitution, constitutes sustained drug-free remission. Drug-free remission does not equal cure, as relapses can still occur after 6 months of follow-up. So far, relapses have been observed within the first two years after the treatment in up to 10% of patients. Long-term drug-free remission may indeed signify cure of AID. While there is no generally agreed definition of cure in relation to AID, 5 years without disease symptoms in the continued absence of immunosuppressive drug therapy would align with the concept of cure in oncology—where 5-year survival rate is a commonly used benchmark by the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Program (https://seer.cancer.gov/).

While immune reset is currently not fully defined at the molecular level, a series of markers may indicate its presence. Complete B cell depletion in tissues appears to be an important marker for sustained drug-free remission. While most of the B cell depletion approaches—including monoclonal antibodies, T cell engagers and CAR T cells—efficiently deplete circulating B cells, their efficacy to deplete B cells within tissues varies[71]. Analysis of lymph nodes of patients with AID treated with B cell depleters showed that full depletion of B cells in the lymphatic tissue was associated with sustained drug-free remission, while partial depletion was associated with recurrence of disease[72]. Full clearance of B cells from lymph nodes also leads to the disruption of germinal centers, with disappearance of follicular dendritic cells and follicular helper T cells—suggesting qualitative changes to lymph node architecture that impair antigen presentation function, with broad effects on the immune cell landscape[66,67].

Indeed, functional analysis of T cells and myeloid cells in patients with SLE after CAR T cell therapy revealed a downregulation of genes associated with type I interferon response, Fc-receptor, complement activation and Toll-like receptor signaling, indicating a deeper revamping of the immune landscape[73,74]. In addition, the numbers of memory B cells and plasmablasts decreased in the blood of patients with AID who received CD19-CAR T cells, while the recurring B cells are naive non-class switched cells[16,17]. B cell receptor (BCR) sequencing of these cells showed the preferential usage of δ- (IgD) and μ- (IgM) heavy chains, indicating a non-class-switched B cell compartment, broad V-J rearrangements without clonality similar to healthy individuals, and downregulated expression of distinct BCR chains associated with autoimmunity[17]. This rebooting of the B cell compartment with loss of plasmablasts after CD19-CAR T cell therapy is associated with the declining levels of most autoantibodies, but not vaccine-induced antibodies. This observation suggests that many autoantibodies may emerge from extrafollicular B cell responses, which create CD19-expressing short-lived plasmablasts rather than long-lived plasma cells. Notable exceptions include anti-SS-A/Ro autoantibodies, which remain stable after CD19-CAR T cell therapy and seem to be derived from long-lived plasma cells, which are CD19-negative and escape CD19-CAR T cell killing. Whether the remaining presence of autoantibodies after CAR T cell therapy constitutes a risk factor for relapses is unclear.

Overall, immune reset after CAR T cell-mediated full B cell depletion is characterized by sustained drug-free remission and a

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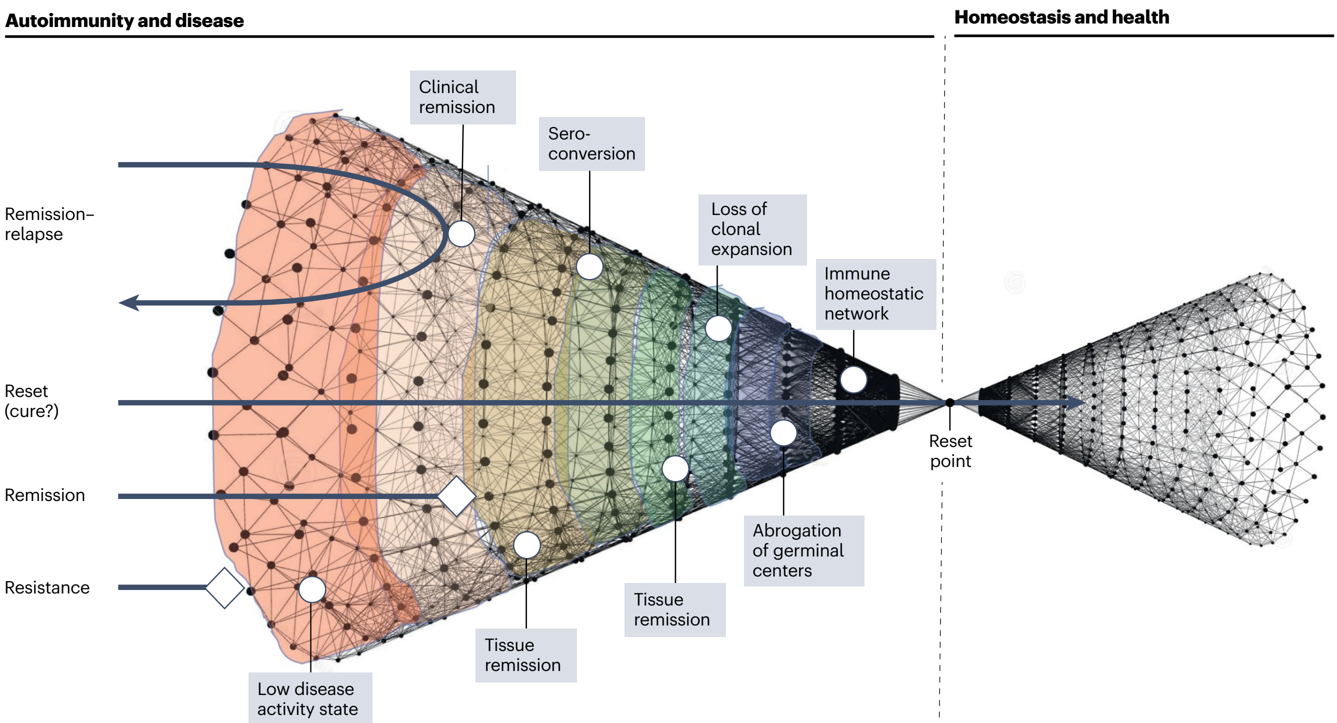

Fig. 3 | Double-cone model of immune reset in AID. The left cone symbolizes autoimmunity and disease, which is shrinking the further one goes to the right with a point of 'singularity' at the reset point; the right cone symbolizes homeostasis and health with no further need for immunomodulatory treatments. To reach immune reset, several checkpoints are required to be passed consisting of shallower (left; for example, reaching a state of low disease activity) and deeper (for example, reaching an immune homeostatic network with normalization of B cell, T cell and myeloid cell activation) components. Arrows from top to bottom indicate different scenarios for autoimmune patients with remission followed by relapse, immune reset and potentially cure, drug therapy-controlled remission and resistance.

'revamping' of the immune landscape, according to endpoints such as immune reconstitution, susceptibility to infections and long-term immune competence.

Challenges of CAR T cell therapy in AID

The complexity of autologous CAR T cell therapy—requiring leukapheresis, manufacturing and in-patient monitoring during conditioning and therapy infusion—make this a resource-intensive and cost-intensive treatment that requires appropriate patient selection. The process of CAR T cell manufacturing has been standardized, making the treatment accessible either through commercial products or through hospital exemption models that allow certified hospitals to develop and use their own approved CAR T products on-site. In this way, several tens of thousands of patients under oncology care have received CAR T cell treatment so far. However, the scalability and costs of autologous CAR T cell therapy would not allow widespread treatment of patients with AID, making careful selection of patients crucial. In this context, patient heterogeneity needs to be considered and the discovery of biomarkers that predict response or relapse needs to be prioritized.

Shortening production time, point-of-care production, reduced doses of lymphodepletion regimens and outpatient administration of CAR T cells may further facilitate its use in the future. Further improvement in scalability may emerge from allogeneic CAR T cell therapy, allowing the use of a single CAR T cells product for multiple patients, avoiding leukapheresis and thereby reducing the overall costs and complexity. Costs accrue at the beginning of CAR T cell therapy but substantially decrease thereafter owing to lack of immunosuppressive treatments and lower secondary costs relating to hemodialysis, kidney transplantation, oxygen treatment and rehabilitative measures, to name a few. As medical costs drop by more than 90% after CAR T cell treatment[75], it may become cost-effective after several years of sustained drug-free remission. Access to CAR T cell therapy is still limited as the treatment requires dedicated centers with disease specialists and cellular therapy specialists working together. On the other hand, CAR T cell therapy is a one-time therapy approach that can be planned with referral of the patients to the specialty care centers for the treatment, while further follow-ups can happen at the local treating physician.

Future directions

Treatment of AID with engineered cells is a rapidly emerging field. In vivo CAR T cell therapy, where targeted lipid nanoparticles (LNPs) carrying mRNA or a viral vector encoding the CAR are infused into patients, holds much promise (Fig. 2). An antibody, for example, against CD5 or CD8, is generally attached to the LNP to specifically direct it to target T cells. Upon injection, the LNP fuses with the respective target cells, releases the CAR payload, which is then translated and allows expression of the CAR in vivo. First experiments in nonhuman primates involved a CD8-targeted LNP containing a CD19-CAR mRNA, and showed CAR expression directed to $CD8^+$ T cells alongside B cell depletion in the circulation and lymphatic tissues[76]. First data in humans using a similar approach have shown rapid expression of the CAR on T cells, B cell depletion and improvement of disease activity in SLE[77]. The feasibility of in vivo CAR T cell therapy in humans has also been supported by data obtained in patients with multiple myeloma who received a lentiviral-based BCMA-CAR that was packed in vesicular stomatitis virus with T cell-targeting properties[78]. The depth of B cell depletion reached by in vivo CAR T cell approaches and their potential to induce immune reset and sustained control of AID are unknown. However, based on the ease of this approach with no necessity of leukapheresis, ex vivo CAR T cell manufacturing or conditioning treatment, there is considerable interest in exploring it. Of note, a one-time

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intervention leading to immune reset may not be the only scenario for in vivo CAR T cell therapy. Other endpoints such as 'immune dimming'—where in vivo CAR T cell therapy is used as a more prolonged immunotherapy approach—will need to be considered if only partial tissue B cell depletion can be achieved. Similar concepts may also apply for T cell engager therapy in AID.

Another new development is the use of 'armed' or fourth-generation CARs, whereby CAR activation is linked to the production of cytokines (for example, IL-15 to enhance CAR T cell expansion) or the production of cytokine-inhibiting antibodies (for example, anti-TNF or anti-IL-6 receptor to neutralize inflammation)[79,80]. These approaches aim to enhance the efficacy of CAR T cell treatment. Hence, more robust CAR T cell expansion owing to a favorable cytokine milieu could allow better B cell depletion, while the production of cytokine-neutralizing antibodies could block the inflammatory component of the disease. Such an approach has been tested in rheumatoid arthritis, a disease that is characterized by a synergy of pathological B cell activation and secretion of proinflammatory cytokines by resident stromal cells and myeloid cells[79].

Selective depletion of antigen-specific B cells (rather than global B cell depletion) is another new approach that offers the possibility for a clone-directed precision therapy. Such an approach may be possible if the exact pathogenic antigen is known, such as desmoglein-3 in pemphigus vulgaris or acetylcholine esterase in myasthenia gravis. To accomplish selective elimination of antigen-specific B cells, CAR cells express the antigen rather than the antibody, which allows binding to the specific B cell clone expressing the matching B cell receptor (chimeric autoantibody receptor cells)[81]. Activity of such cells, however, may be neutralized by autoantibodies binding to the chimeric autoantibody receptor cells, preventing them from binding to the BCR. Another concept for clone-directed cellular therapies is to direct CARs against certain disease-associated BCR chains, like the immunoglobulin heavy variable gene (IGHV) 4-34, which is associated with SLE. IGHV 4-34 + B cell clones are increased in SLE, and IGHV 4-34 chains are found in SLE-associated autoantibodies, such as anti-dsDNA[82]. CARs specifically targeting IGHV 4-34 could thus selectively eliminate pathogenic B cell clones without triggering full B cell depletion. While such CAR T cells have been shown to selectively kill respective B cell clones in vitro, the in vivo efficacy of such an approach in preclinical models of SLE and patients with SLE remains to be shown[83].

CD19 and BCMA serve as the prime targets for CAR T cells in AID. In diseases in which processes other than B cell activation prevail, the technology of engineered cells may provide additional targets for intervention. For instance, fibroblast activation protein, expressed on activated fibroblasts in the synovial membrane of rheumatoid arthritis as well as in the skin and lungs of patients with SSc, could be a target for CAR T cells. Fibroblast activation protein-specific CARs have so far been used only in preclinical models of myocardial fibrosis after infarction, showing reduction of fibrotic tissue remodeling[84]. The safety and efficacy of such an approach in humans is unknown. Apart from that, T cell targeting by CAR T cells may allow the selective elimination of a pathogenic T cell subset as broad depletion of T cells is not compatible with safety. Potential approaches include the targeting of certain disease-associated TCRs such as TRBV9, which is associated with axial spondylarthritis and which is only expressed in a subset of T cells[85].

Finally, CAR regulatory T (T_{reg}) cells may have a future role in the treatment of AID. In contrast to CAR T cells, these cells do not kill cells but promote a regulatory homeostatic environment[86]. In contrast to non-engineered polyclonal T_{reg} cells, which exert only time-limited effects in the absence of sustained activation signals, CAR activation in engineered T_{reg} cells may provide longer stability as well as proliferation of these cells. So far, preclinical efficacy of CD19-CAR T_{reg} cells in SLE has been reported[87]. Clinically, preliminary data have been reported on CAR T_{reg} cells recognizing citrullinated proteins, which aim to induce a local immunoregulatory environment at sites of protein citrullination in patients with rheumatoid arthritis[88]. Further studies on CD19-CAR T_{reg} cells in AID are currently in development, but no clinical data have yet been reported.

Conclusion

Within a few years, CAR T cell therapy has advanced our understanding of AID and driven long-term drug-free remission in patients with severe disease—an unexpected finding that led to the concept of 'immune reset', a fundamental revamping of the pathological immune landscape. The rapid increase in clinical studies of CAR T cell therapy in AID combined with fast technological advances makes this field highly dynamic, providing entirely new possibilities to manage AID in the future. These developments are flanked by a reconceptualization of clinical trials in AID and a fundamental change in the concept of treatment objectives, shifting from long-term immunosuppression to short-term treatment and durable immune reset, with potential even for cure.

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Author contributions

G.S. and H.X. wrote the draft version and G.S. revised the manuscript.

Competing interests

G.S. received speaker honoraria from BMS, Cabaletta, Eli Lilly, Johnson & Johnson, Kyverna, Novartis, Orbital and UCB. H.X. declares no competing interests.

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